

Isolated Level-shifting in Monolithic GaN

Alen Komadina

Thesis submitted for the degree of
Master of Science in
Electrical Engineering, option
Electronics and Chip Design

Supervisor

Prof. dr. ir. Filip Tavernier

Assessor

Prof. dr. ir. Tim Piessens

Assistant-supervisor

Ir. Öncü Benli Marin Palomo

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Published by Alen Komadina,
Department of Electrical Engineering, Kasteelpark Arenberg 10 bus 2440, B-3001 Leuven

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Preface

It's always darkest before the dawn.

And in that sense, this is my acknowledgment to thank everyone who provided me with a challenging, yet exhilarating working environment, and to everyone who helped me bear the countless hours while working on this thesis.

I would like to thank my promotor, prof. dr. ir. Filip Tavernier, and Dr. ing. Jef Thoné, for providing an opportunity to collaborate with MinDCet NV on such an interesting and relevant topic. Furthermore, I owe endless gratitude to my daily supervisor, Ir. Öncü Benli Marin Palomo, for continuous support and countless words of advice during the past academic year. I had the pleasure of learning so much from you, and I hope I managed to show it in this work. Also, I am grateful to the entire jury for taking the time and evaluating the thesis.

Naturally, an incandescent *Thank you* to my family and friends who were there from the very first day, listening and giving me the necessary pushes I needed to be able to finish any objective, especially this work. Among them, a special thanks to my dear friend Dorotea - I am sorry for all the rant bridges, they were sadder than T. S. songs. Finally, an insurmountable thanks to my mother - without your wisdom and support, there would be nothing.

Alen Komadina

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Abstract

This thesis focuses on the concept development - IC design and simulation - of two galvanically isolated, digital level shifter topologies in monolithic GaN. The level shifter is aimed to be used as a part of a gate driver for a half-bridge. Both topologies are compared in terms of attainable Common-Mode Transient Immunity (CMTI), propagation delay, current consumption, and estimated area. The two topologies use different methods to handle CMTI, the details can be found in [1] and [2]. The IC design of the topologies in this paper differs slightly due to a lack of d-mode HEMT's in the PDK used.

Both level shifters need to translate the voltage from the low-side domain (0 - 6 V) to the high-side domain (80 - 86 V), while achieving sufficient CMTI - $|dV_{SSH}/dt| > 100 \text{ V/ns}$ - and sufficiently low propagation delay - $t_d < 10 \text{ ns}$. The circuits are implemented in imec's gan200 technology, and their operation was simulated up to 2 MHz.

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List of abbreviations and symbols

Abbreviations

2DEG	Two-Dimensional Electron Gas
CMBP	Common-Mode Blanking Pulse
CMDC	Common Mode Displacement Current
CMTI	Common Mode Transient Immunity
DDC	Dynamic Discharge Control
FBDC	FeedBack Discharge Control
FLB	Fault Logic Blanking
FoM	Figure of Merit
GaN	Gallium Nitride
HEMT	High Electron Mobility Transistor
IC	Integrated Circuit
LS	Level Shifter
MMRC	MisMatch Reduction Circuit
MOSFET	Metal-Oxide-Semiconductor Field Effect Transistor
NF	Number of Fingers
OCC	Operating Condition Check
PDK	Process Design Kit
PEX	Parasitic Extraction
SiC	Silicon Carbide
SNFED	Switching Node Falling Edge Detection
SRFF	Set-Reset Flip-Flop
WF	Width of a Finger

Symbols

A	Area
E_{crit}	Critical (Electric) Field
E_g	Band Gap
ε_r	(Relative) Permittivity
f_{sw}	Switching Frequency
I_{DC}	DC Current Consumption
I_X	Current through Wire X
L_X	Length of Transistor X
λ	Thermal Conductivity
μ_n	Electron Mobility
π	Number Pi
Q_{gate}	Gate Charge
$R_{DS,on}$	On-resistance
t_{fall}	Fall Time
t_{delay}	Delay Time
t_{rise}	Rise Time
V_X	Voltage at Node/Terminal X
W_X	Width of Transistor X

Chapter 1

Introduction

The *Introduction* contains a general summary of the work done - what are the goals to be achieved, what is the motivation behind using Gallium Nitride as a semiconducting material for this work, and how the desired goals may be reached provided a certain Gallium Nitride technology.

1.1 Problem Definition

In order to simultaneously support the increase in society's electrification and the decrease in carbon emissions, more efficient power conversion technologies are needed. For this purpose, wide-bandgap semiconducting materials such as Silicon Carbide (SiC) and Gallium Nitride (GaN) have been introduced to produce power transistors.

In addition, GaN has lower C_{ISS} when compared to Si, which means it can switch faster. High switching speeds lead to fast common-mode transients, which pushes the need for extremely robust level shifting techniques to safely translate signals from low-voltage levels to high-voltage levels. Moreover, the application in question requires galvanic isolation via e.g. capacitors or transformers - but following the trend of monolithic GaN integration, these requirements pose quite a challenge. In this thesis, the described challenge is tackled.

1.2 Why GaN?

Before any design is performed, one might pose the question: "Why use a compound such as GaN if there is an abundance of cheap, good-quality Silicon available?".

As the level shifter must operate in the voltage range of 0 - 6 V, the devices used in the level shifter topologies must have high enough breakdown voltage. The wurtzite structure (see Appendix A for more details) of GaN crystal lattice has stronger chemical bonds, leading to the higher critical electric field, and thus the higher breakdown voltage of GaN-based device compared to the breakdown voltage of an Si-based device.

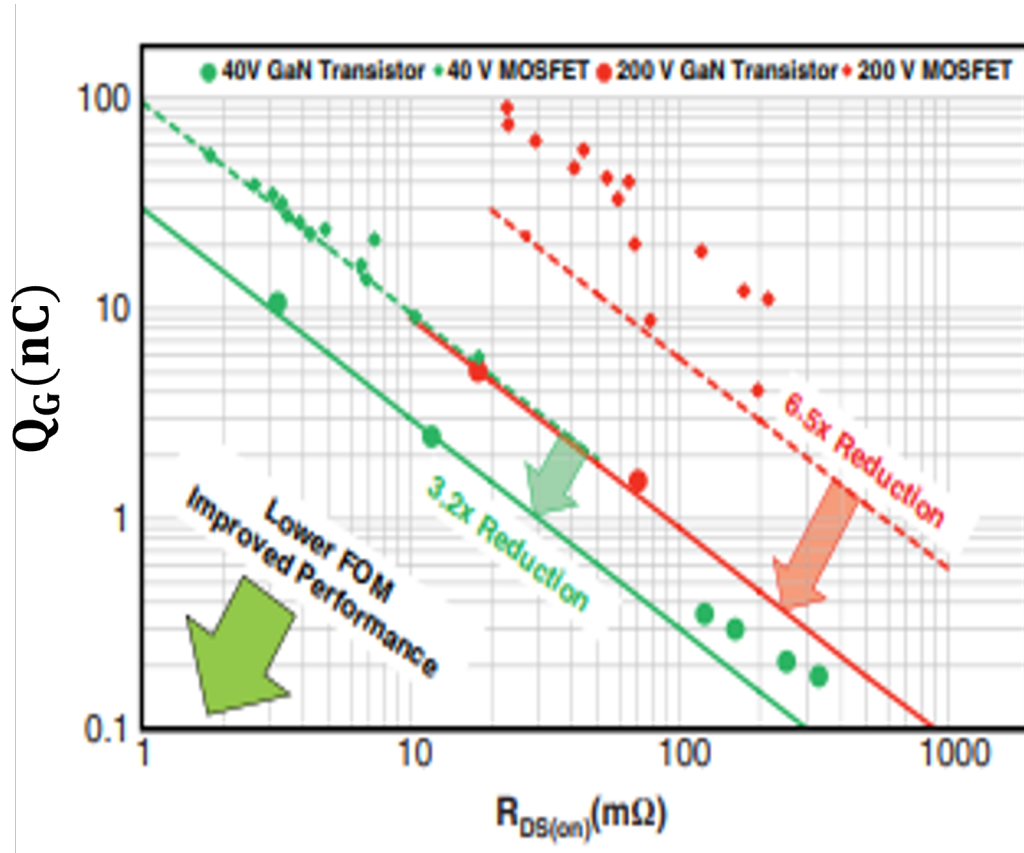


FIGURE 1.1: "On-resistance vs. total gate charge comparison for Si- and GaN-based 40 V and 200 V power devices." [3]

Furthermore, regarding the performance comparison between GaN- and Si-based devices, two important figures of merit (FoM's) must be taken into account:

1. $R_{DS,on} \cdot Q_{gate}$ - for the same on-resistance, GaN exhibits lower gate charge than Si (shown in Figure 1.1), making it suitable for faster switching (but this may be a drawback when the IC has a lot of ringing);
2. $R_{DS,on} \cdot Area$ - for the same on-resistance, GaN-based design can be more compact than Si-based design.

1.3 Solution Specifications

Level shifter specifications to be met are given in Table 1.1. As may be seen, the main design constraints include sufficient speed (i.e. the delay must be lower than the values stated in Table 1.1) and the ability to function even when unexpected, rapid voltage signal transients occur (characterized by CMTI - Common Mode Transient

Specification	Value	Unit
Technology	imec GaN 200 V	-
Maximum V_{in}^*	80	V
f_{sw}	< 2	MHz
CMTI	100	V/ns
Propagation delay (pre-PEX)	< 10	ns
Propagation delay (post-PEX)	< 15	ns
Propagation delay matching	< 3	ns
Minimum on-off times	25	ns
Maximum on-off times	500	ns

TABLE 1.1: Level Shifter Specifications.

Immunity - up to 100 V/ns). After satisfying these initial constraints, optimizations such as power and area management can be executed.

1.4 Workflow

As a first step towards reaching the defined goal, a literature study is needed. The designer should understand not only the motivation behind the use of GaN as a semiconducting material but also the operating principle of GaN-based transistors. After grasping the fundamentals, a study on circuit topology of the level shifter may be done - which topology would best suffice for given requirements, and which potential changes to it are needed.

When the appropriate topology is selected, the actual concept development may begin; circuit design and layout may be executed. After the circuit layout has been implemented, post-layout simulations are to be executed to further verify the functionality of the proposed design. After this step, the circuit is ready for fabrication and measurements.

In the following document, solution steps up to (and excluding) layout design are elaborated on for two level-shifting topologies, which are then compared in terms of speed (i. e., propagation delay requirements), CMTI limits, total DC-current consumption, and estimated area.

Chapter 2

Literature Study

In the following chapter, a brief description of a GaN transistor working principle is given, as well as an introduction to a general level shifter. Starting from this chapter, and excluding the final chapter (Chapter 5), a short summary will be given at the end of each chapter to emphasize the most important concepts used and the results achieved.

2.1 GaN Transistors

2.1.1 The Two-Dimensional Electron Gas

GaN exhibits an interesting hexagonal, crystal structure called wurtzite (see Appendix A for more details). This structure gives GaN piezoelectric properties which lead to it being able to achieve very high conductivity compared to other semiconductors [3].

If the lattice is subjected to strain, its deformation will lead to an electric field generation proportional to the strain applied. To achieve this, a thin layer of AlGaN on top of the GaN crystal may be grown (as shown in Figure 2.1). At the interface of these two layers, a so called two-dimensional electron gas (2DEG) will be induced. Hence, a reservoir of highly conductive, confined electrons is formed, providing a basis for the high electron mobility transistor (HEMT) [3], the primary component used for the design described in this work.

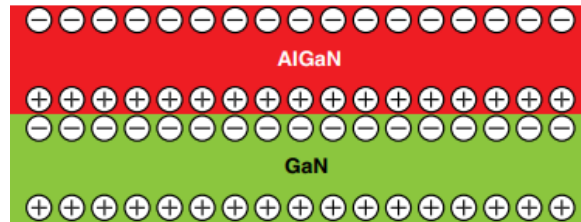


FIGURE 2.1: "Simplified cross section of a GaN/AlGaN heterostructure showing the formation of a 2DEG due to the strain-induced polarization at the interface." [3]

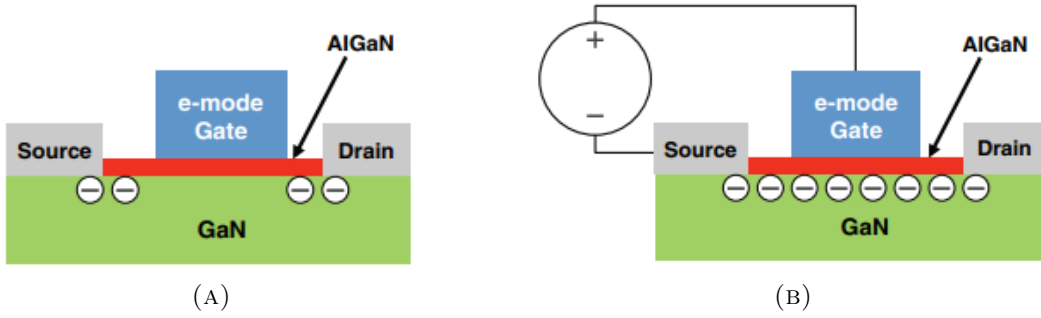


FIGURE 2.2: (A) "An enhancement-mode (e-mode) device depletes the 2DEG with zero volts on the gate." [3] (B) "By applying a positive voltage to the gate, the electrons are attracted to the surface, re-establishing the 2DEG." [3]

2.1.2 The Basic GaN Transistor Structure

As with any Field Effect Transistor (FET), HEMT's are comprised of four electrodes: bulk, drain, gate and source electrode. The source and drain electrodes pierce through AlGaN to make an ohmic contact with the 2DEG and allow for a current flow through the transistor [3]. In order to control this current flow, a gate electrode is placed on top of the AlGaN layer [3]. A side-remark worth mentioning is that the process used in this work is GaN-on-Si process, and all the bulks are connected to the same node within the same chip. This means that level shifters are split into two different chips based on the voltage domain due to the bulk connections.

There are two main types of HEMT's – a depletion-mode and an enhancement-mode transistor. The difference is as follows (paraphrased from [3]): the depletion-mode HEMT conducts current when a zero bias is applied to its gate; in other words, it turns off only when a negative bias gate voltage is applied. On the other hand, the enhancement-mode HEMT behaves in such a way that it is off when the bias gate voltage is zero, and it conducts current when a positive bias gate voltage is applied (as illustrated in Figure 2.2). The technology node used in this project is restricted to the enhancement-mode HEMT's; therefore, only this type of transistor will be referred to in further text and figures.

2.2 Level Shifters

A level shifter is an electronic circuit used to translate a signal from one voltage level to another, such that devices with different supply voltages may communicate safely. In essence, it is used to bridge different voltage domains, and ensure that both low-side and high-side components remain functional as well as robust to any damage.

Level shifters are commonly used between a micro-controller and sensor interfaces, in communication busses (I^2C , SPI, UART), in memory cards and embedded

devices, but also in power electronics' gate drivers - the application this work focuses on.

One may categorize level shifters in various ways, e. g.:

- Direction-based classification: if the signal flows only one way (from input to output), a level shifter is unidirectional; on the contrary, if the signal flows in both directions (connected devices transmit and receive), a level shifter is bidirectional. This work focuses on a unidirectional level shifter design.
- Implementation-based classification: simple level shifters might be designed on the basis of resistive division or MOSFET-based topology, while more complex level shifters are dedicated integrated circuits comprised of a low-side part and a high-side part with a capacitive/transformer isolation. The topologies used in this work will be elaborated in Chapters 3 and 4.

Based on design requirements, an optimal level shifter topology is to be selected and implemented. For example, if a high-to-low level signal translation is needed, where the primary concern is to have a simple and small component without much isolation between voltage domains, a resistive divider may be used as a level shifter. If, however, a high-speed, high CMTI (well isolated), low-to-high level shifter is needed, a much more complex topology must be designed to meet all the specifications. Such a challenge is tackled with in this work - design steps and choices made are thoroughly explained in following chapters.

2.3 Summary

Understanding the operation of a single GaN-based transistor will greatly help to design a complex level shifting system at hand. That is why in this chapter fundamental principles of GaN transistors were explained - how the strained crystal GaN structure enables a highly conductive 2D Electron Gas to be formed at the interface between GaN and AlGaN layers, leading to the development of a High Electron Mobility Transistor [3], the primary component used in this work.

Moreover, a short introduction to the concept of level shifter circuits was given to clarify the general idea behind them - what a level shifter is, what its applications are and how it can be characterized.

Chapter 3

Circuit Design - Topology 1

Now that one has a good idea of which components are to be used in which kind of circuit, it is time to move on to the actual design. In the following section, the first proposed level shifter topology and operation principle are described. Afterwards, the design methodology of the circuit is elaborated - starting from simple building blocks such as logic gates, and towards the entire system. In addition, relevant simulation results are presented and discussed.

3.1 Proposed Level Shifter and Its Operation Principle

The first proposed level shifter topology based on [1] is shown in Figure 3.1 and is discussed in this section.

As stated in [1] and seen in Figure 3.1, the topology in question utilizes two techniques which work hand-in-hand to prevent the overcharge of the high-side power supply capacitor during high (negative) dV/dt transitions and improve CMTI. **Note:** in the following paragraphs of this section, all the signals, nodes, and components mentioned are the ones seen in Figure 3.1.

The first technique employed is the so-called Switching Node Falling Edge Detection (SNFED). In short, the operating principle is as follows: during the falling edge of V_{SSH} , the common mode displacement current $I_{CMD C}$ is generated and flows through C_B , generating an overshoot at N_3 (N_4). The sense blocks (inverters) detect this change, and the SNFED-block generates a blanking signal V_B to suppress the faulty logic signal by turning on the transistor E_7 (E_8), such that $I_{CMD C}$ flows safely to V_{SSH} . [1] may be consulted for more details.

The main bottleneck of SNFED is the turning speed of E_7 (E_8), which is limited by the propagation delay of the digital loop [1]. This means that SNFED does not entirely help in the case of high dV/dt transitions (meaning $50V/ns$ and above, as discussed in [1]), such as in half-bridge GaN drivers where the level shifter from this work is to be used. That is why a second optimization is needed, namely the Dynamic Discharge Control (DDC) circuit.

DDC operates on the following principle: when a high negative dV_{SSH}/dt occurs, $I_{CMD C}$ is generated and injected to the high-side part via capacitors C_{SENSE} . This

3. CIRCUIT DESIGN - TOPOLOGY 1

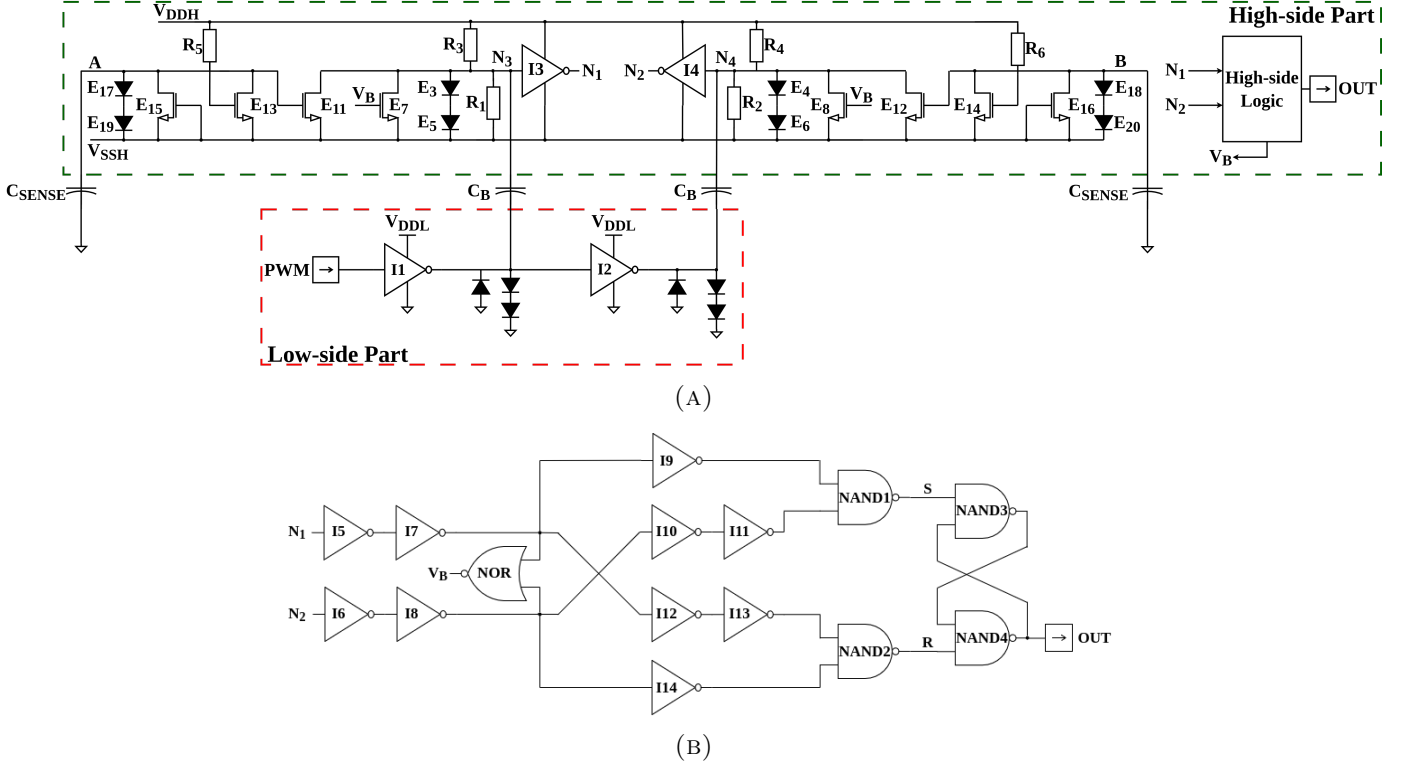


FIGURE 3.1: (A) Capacitive level shifter topology based on [1]. (B) High-side logic comprised of the Fault Logic Blanking (FLB) and a Set-Reset Flip-Flop (SRFF), respectively.

current will normally flow through diodes $E_{17,19}$ ($E_{18,20}$) and transistor E_{13} (E_{14}), and the voltage V_A (V_{nodeB}) will rise such that transistor E_{11} (E_{12}) turns on and enables the discharge of C_B to V_{SSH} . In the case of positive dV_{SSH}/dt , C_{SENSE} is charged by the current path through the transistor E_{15} (E_{16}). [1] may be consulted for more details.

A few notable changes with respect to the topology described in [1] are:

- diodes $E_3 - E_6$, which replace transistors $M_{5,6}$ from Figure 6 in [1] - in this way, nodes N_3 and N_4 are not pulled low constantly, but only after the voltage peaks needed for sensing inverters occur (otherwise the circuit malfunctions);
- resistors R_1 and R_2 are added to ensure a high enough voltage value and a long enough RC-discharge constant τ , such that sensing inverters are able to capture any change at nodes N_3 and N_4 ; resistors $R_3 - R_6$ are added after the execution of Operating Condition Check (OCC; explained in the last section of this Chapter);
- transistors M_9 and M_{10} from Figure 6 in [1] are omitted since they do not contribute to this topology;
- diodes in the low-side part are added to sink I_{CMDC} during high positive dV/dt and ensure that the voltages over low-side transistors do not cross the

(A)			(B)		
Parameter	Value	Unit	Parameter	Value	Unit
R	915	Ω	C_b	25.6	fF
W_1	10	μm	W_1	2	μm
			W_2	2	μm
			W_3	4	μm

TABLE 3.1: Sizing of the inverter (A) *Type 1*, (B) *Type 2*.

breakdown threshold.

In the following sections, elaborate design steps and simulation results obtained are discussed.

3.2 Logic Gates Design

As is the case with the majority of the level shifter topologies, topologies used in this work incorporate logic gates such as inverters, NAND-gates, and NOR-gates. Hence, the first step towards a fully functional level shifter topology is designing sufficient logic gates. In the following subsection, the design considerations and choices for the inverter logic gate are discussed.

3.2.1 Inverters

Taking into account the limitation of the technology used in this work, that is, not having a p-type e-mode HEMT, one might intuitively opt for the simplest possible inverter network, where the pull-down network consists of a single n-type HEMT, and the pull-up network is a resistor. This topology is shown in Figure 3.2a and from now on, it will be referred to as the *Type 1* inverter. Area-wise, this design is extremely compelling if the resistance needed is low enough, but it might not be as feasible regarding its operation speed and power consumption.

Therefore, an optimized topology is considered, as depicted in 3.2b and based on [4], but slightly modified. This *Type 2* inverter works the following way: the pull-down network naturally switches on when the input voltage V_{in} is high, but in the meantime the diode-connected E_1 charges the capacitor C_b . When V_{in} switches from high to low, then $V_{G2} = 2V_{DD} - V_{TH1}$ holds due to the previously charged capacitor. From this, it follows $V_{GS2} = 2V_{DD} - V_{TH1} - V_{OUT}$, which ensures that E_2 can pull V_{out} high when V_{in} is low.

The two described topologies were simulated and the results were compared bearing in mind that the main specification to be met in the overall design is timing. Both were initially sized referring to [4], and then modified to meet desired specifications (for example, if the pull-down network was too weak (i. e. it has too high of a resistance), its width was increased twice to make it twice as stronger). All transistors $L_{min} = 1.3 \mu m$ as the PDK allows for only a fixed gate length, and other parameters are given in Table 3.1.

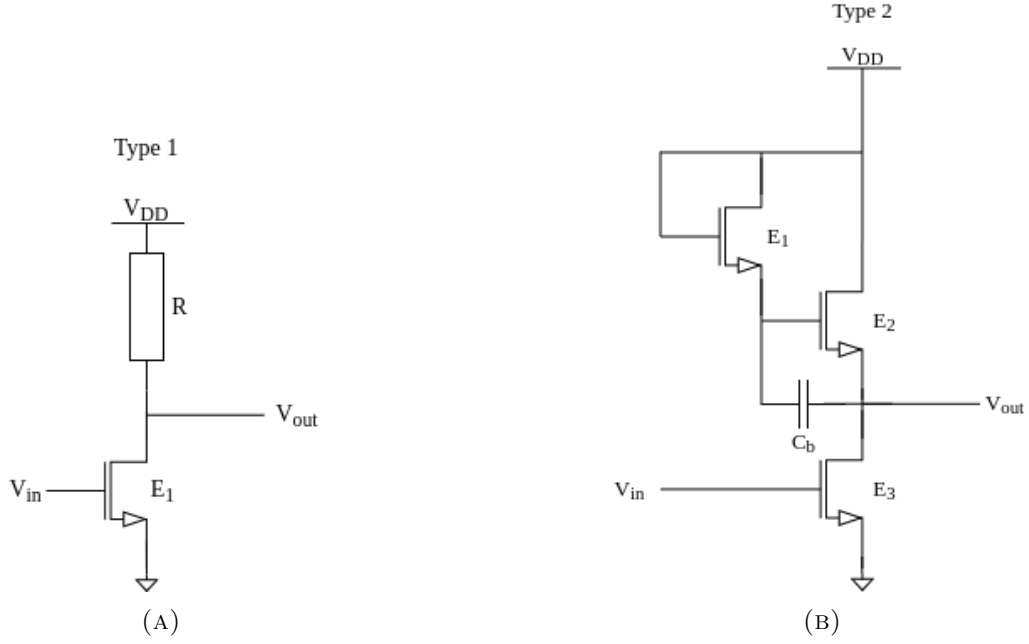


FIGURE 3.2: Inverter topologies to be compared, where the pull-up network is implemented by means of a (A) resistor, (B) charge-pump circuit [4].

Specification	Value (<i>Type 1</i>)	Value (<i>Type 2</i>)	Unit
t_{rise}	1.35	0.92	ns
t_{fall}	1.03	1.09	ns
t_{delay}	1.19	1.01	ns
I_{DC}	363.54	316.11	μA
A_{est}	5730	2310	μm^2

TABLE 3.2: Simulation results for inverter topologies.

The transient analysis waveforms are given in Figure 3.3 - the analysis was done at the frequency of 2 MHz, with a load capacitance of 100 fF (more than the expected load for a chain of logic gates). It may be seen that both topologies perform their intended function. The results needed for comparison are given in Table 3.2. As may be observed, the propagation delay and current consumption are somewhat lower (around 0.2 ns, 50 μA , respectively) for the *Type 2* inverter than for the *Type 1* inverter. Moreover, the estimated area is lower for the *Type 2*, which is due to overhead introduced to the resistor in the *Type 1* - the minimal value of the resistance allowed by the technology used is too high, so multiple parallel segments must be used to achieve the same resistance in the pull-up of the *Type 1* inverter as when looking into the pull-up of the *Type 2* inverter. For these reasons, at this stage the *Type 2* inverter is a preferred topology, hence other logic gates will be derived from it. More details on these logic gates are given in Appendix B.

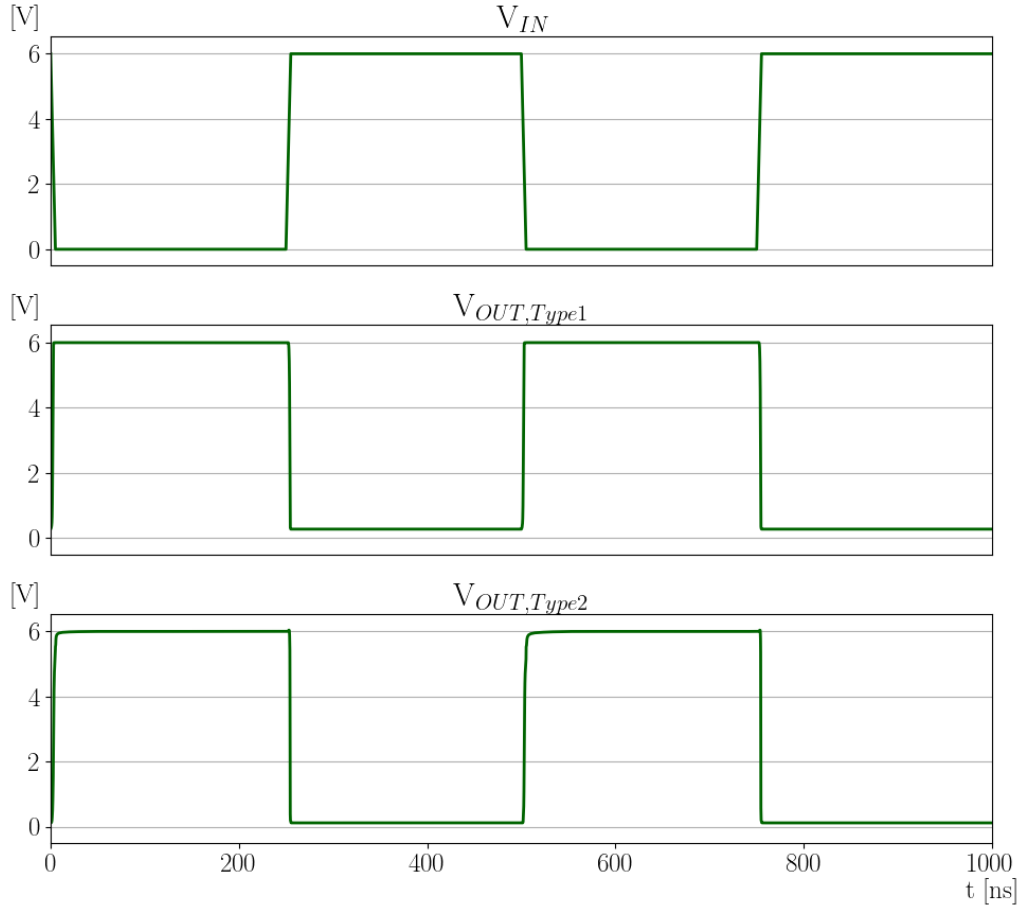


FIGURE 3.3: Transient analysis results for the inverter topologies.

3.3 Level Shifter Design

Now that the basic building blocks are implemented, the remainder of the level shifter circuit will be designed such that all the requirements are satisfied. This section focuses on explaining the sizing rationale of all the circuit components, how the DC, transient, and corner analyses are executed, and how the level shifter is tested in a simulated half-bridge environment.

3.3.1 Sizing the Circuit

In the following subsection, design choices regarding sizing of the components in the capacitive level shifter are discussed. **Note:** for the rest of this chapter, the "level shifter" discussed is the one in Figure 3.1, and in the following paragraphs of this subsection, all the signals, nodes, and components mentioned are the ones seen in Figure 3.1.

Parameter(s)	Value	Unit
C_B, C_{SENSE}	102.4	fF
R_1, R_2	80	$k\Omega$
R_3, R_4	400	$k\Omega$
R_5, R_6	5	$k\Omega$
$W_3 - W_6$	360	μm
W_7, W_8	60	μm
W_{11}, W_{12}	25	μm
W_{13}, W_{14}	80	μm
W_{15}, W_{16}	10	μm
$W_{17} - W_{20}$	120	μm
W_{LS}^*	400	μm

TABLE 3.3: Sizing of the first capacitive level shifter topology. $^*W_{LS}$ refers to widths of the low-side diodes.

All capacitors were sized according to [1], with slight value reduction after transient analysis simulations.

As for diodes, the fact that they must sink $I_{CMDC} \sim 10 - 20 \text{ mA}$ leads to the need for devices of large width. The classical current-model equation (Equation 3.1) is used to derive the needed width of clamping diodes. μ, C_{ox}, V_T are extracted from the PDK, while the assumptions were: $L = L_{min}$, $I_{DS} = 20 \text{ mA}$, $V_{GS} = 3.5 \text{ V}$. Similar analysis was done for transistors E_7, E_{13}, E_{15} (E_8, E_{14}, E_{16}).

$$I_D = \frac{\mu C_{ox}}{2} \frac{W}{L} (V_{GS} - V_T)^2 \quad (3.1)$$

To ensure a long enough τ at nodes N_3 and N_4 ($\sim 1 \text{ ns}$) such that sensing inverters may capture V_{N3}, V_{N4} peaks, resistors R_1, R_2 were swept $10 \text{ k}\Omega - 100 \text{ k}\Omega$, and the minimum possible resistance values from this range were chosen. Other resistors were sized to ensure OCC requirements are satisfied, which is discussed in subsection 3.3.4.

Based on the rationale described and after quite a few iterations, the dimensions of all components, excluding previously elaborated logic gates, are presented in Table 3.3. Note that all transistors are of length $L_{min} = 1.3 \mu m$. More detailed sizing of logic gates after the final half-bridge configuration simulations is given in Appendix D.

Regarding logic gates; initially, each logic gate was taken as an instance described in the previous section or Appendix B. During corner analysis (described later on in this Chapter), instances with stronger pull-down networks had to be used in certain parts of the circuit to ensure proper operation at elevated temperatures (up to $125^\circ C$). Moreover, during the power density check (described in the end of this Chapter), some logic gates exhibited too much power ($> 10 \text{ W/mm}^2$) over small transistors (transistors with $W < 5 \mu m$), hence these gates were replaced with *Type 1* logic gates. Detailed sizing for all components is given in Appendix D.

3.3.2 DC Analysis and Transient Analysis

Due to the nature of the level-shifting circuit, the primary focus is on the transient analysis results. Nevertheless, a DC simulation was executed to verify whether each node has a proper starting point (mainly a logical one or a logical zero).

After confirming the DC operating point was valid, the transient analysis is performed. As the circuit operating frequency is $2MHz$, a conservative transient analysis lasts for $1.1\mu s$ to capture two periods of the output signal. The analysis was performed for a maximum absolute dV_{SSH}/dt of $100V/ns$ to ensure that the circuit satisfies given specifications.

As in [1], the most important voltage signals are shown in Figure 3.4. As can be seen, when a low-side current, generated due to the input signal toggling, is injected into the high-side of the level shifter, there is a brief pulse at nodes N_3 and N_4 . This is captured by the sensing inverters, and forwarded to the rest of the logic, where set (S) and reset (R) pulses are generated to toggle the output accordingly. In the case of an unwanted dV_{SSH}/dt , the generated common-mode displacement current is safely sunk to ground without endangering any logic from sensing inverters on, and hence the output result remains correct.

3.3.3 Corner Analysis

As the given process technology is restricted to enhancement-mode HEMT's only, there are fewer corner cases to test than in the case of CMOS technologies. There will hence be typical (t), fast (f) and slow (s) corner. These three cases were simulated at -40 , 27 and $125^\circ C$, resulting in nine corners.

Figure 3.5 shows the output results for the nominal case (t , $27^\circ C$) and the two cases that distinguish themselves the most with respect to the nominal case (corner s , $125^\circ C$ and corner f , $-40^\circ C$). One may observe that the propagation delay minimizes for lower temperatures, while the performance of the circuit somewhat degrades at higher temperatures, especially for the case of slow (s) corner. This can be surmised from the longer propagation delay (all pull-up resistances roughly double at $125^\circ C$ compared to $27^\circ C$), which indicates a slower overall performance of the logic gates throughout the circuit. Figure 3.5 is a zoom-in to the third operating cycle, since the propagation delays stated in this work were calculated during this cycle.

3.3.4 Half-Bridge Configuration

After simulating the level shifter circuit in a CMTI-free environment, the circuit is put into a half-bridge configuration, which mimics a real circuit environment (with $CMTI > 100 V/ns$), where the level shifter should be able to work properly. Such configuration is shown in Figure 3.6 - cap_lvsh_LS corresponds to the **Low-side Part** and capacitors in Figure 3.1, while cap_lvsh_HS corresponds to the **High-side Part** in Figure 3.1. As can be seen, nodes N_3 and N_4 are cross-coupled to ensure the output signal of the level shifter has the same sign as the input signal of the level shifter, because the level shifter initially provides an opposite-sign output

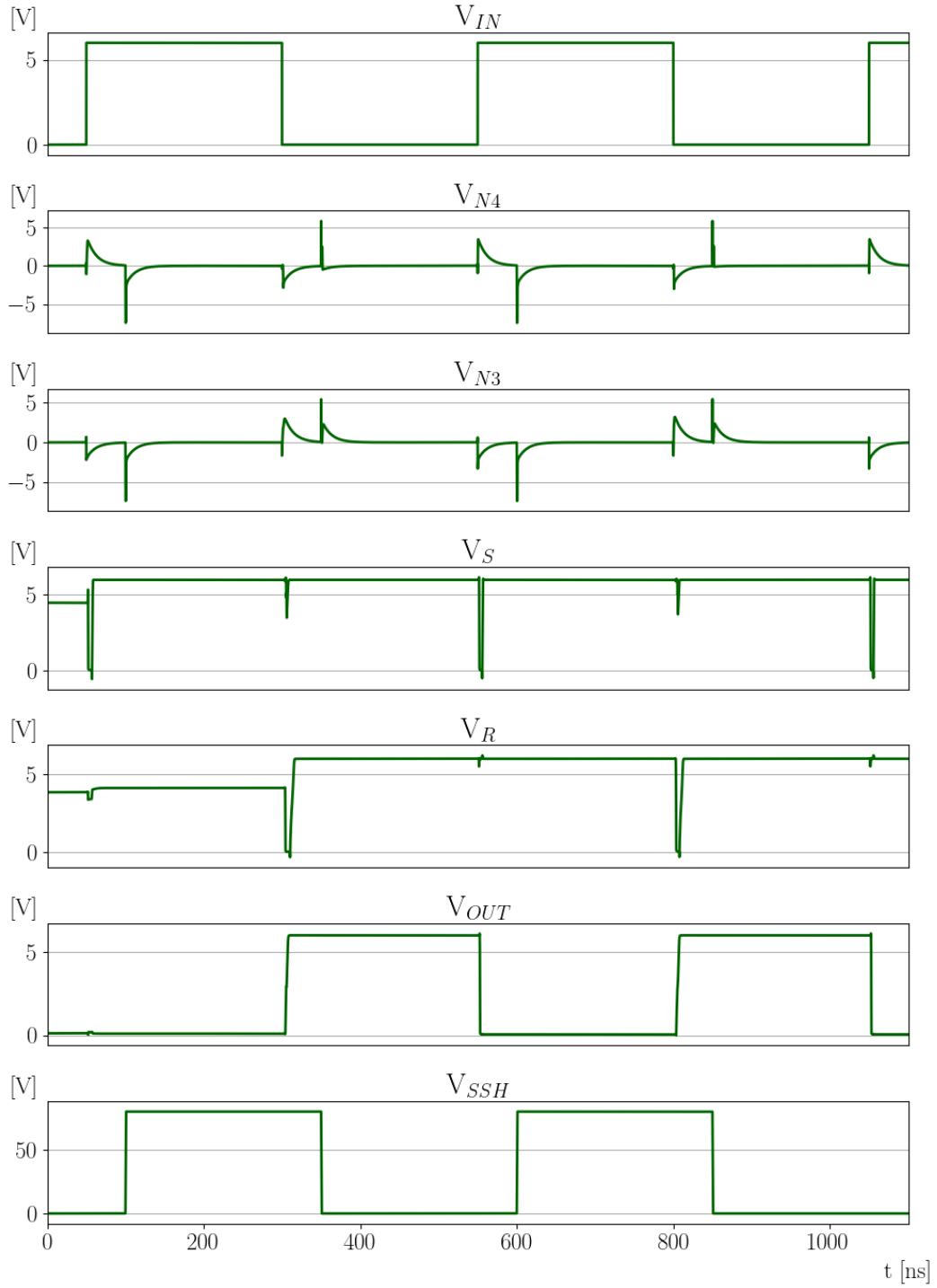


FIGURE 3.4: Transient analysis results for the first capacitive level shifter topology. V_{IN} and V_{SSH} are referenced to (low-side) ground (0V), while V_{N3} , V_{N4} , V_S , V_R and V_{OUT} are referenced to V_{SSH} .

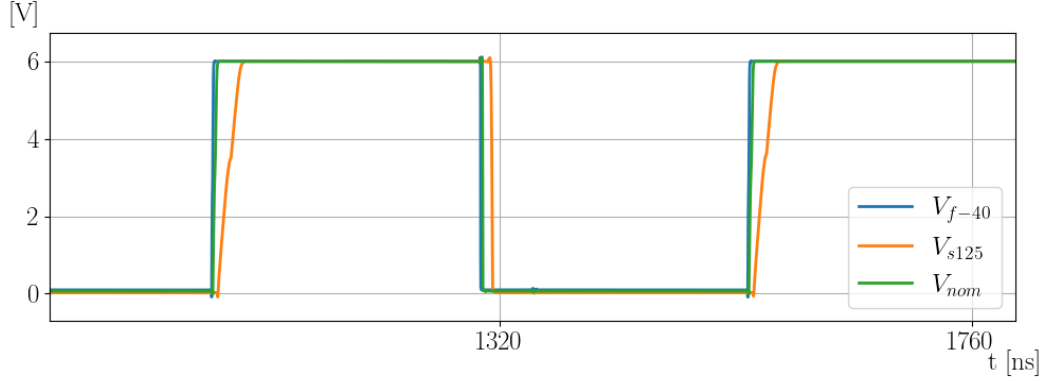


FIGURE 3.5: Corner analysis results for the output of the first capacitive level shifter topology, for the nominal case (V_{nom}) and worst-case corners: s , $125^\circ C$ (V_{s125}), and f , $-40^\circ C$ (V_{f-40}).

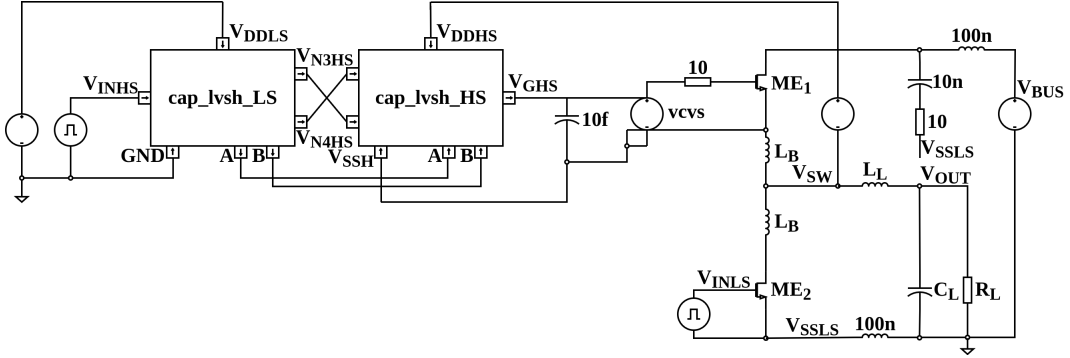


FIGURE 3.6: Schematic of the half-bridge configuration for simulating the level shifter in a real circuit environment.

with respect to the input. **Note:** in the following paragraphs of this subsection, all the signals, nodes, and components mentioned are the ones seen in Figure 3.6.

The primary idea is to examine the influence of the inductive coupling modeled by inductors $L_B = 1 \text{ nH}$ (this value was chosen to prevent the potential for high ringing of the signals V_{SW} and V_{SSH}) on how the floating supply V_{SSH} changes and how it affects the performance of the level shifter. An important note for the input signals' settings has to be realized: voltages V_{INHS} and V_{INLS} must be non-overlapping, otherwise the half-bridge will break down at the moment when both signals are high, i. e. when both ME_1 and ME_2 are turned on due to current shoot-through. Other simulation settings are elaborated in Appendix C.

The results of the transient analysis are given in Figure 3.7, for a larger number of cycles to ensure the circuit is functional. It may be observed that the level shifter does indeed operate as intended - the output signal follows the input signal with a slight delay, and in the event of high dV_{SSH}/dt ($> 100 \text{ V/ns}$), the output signal is not corrupted.

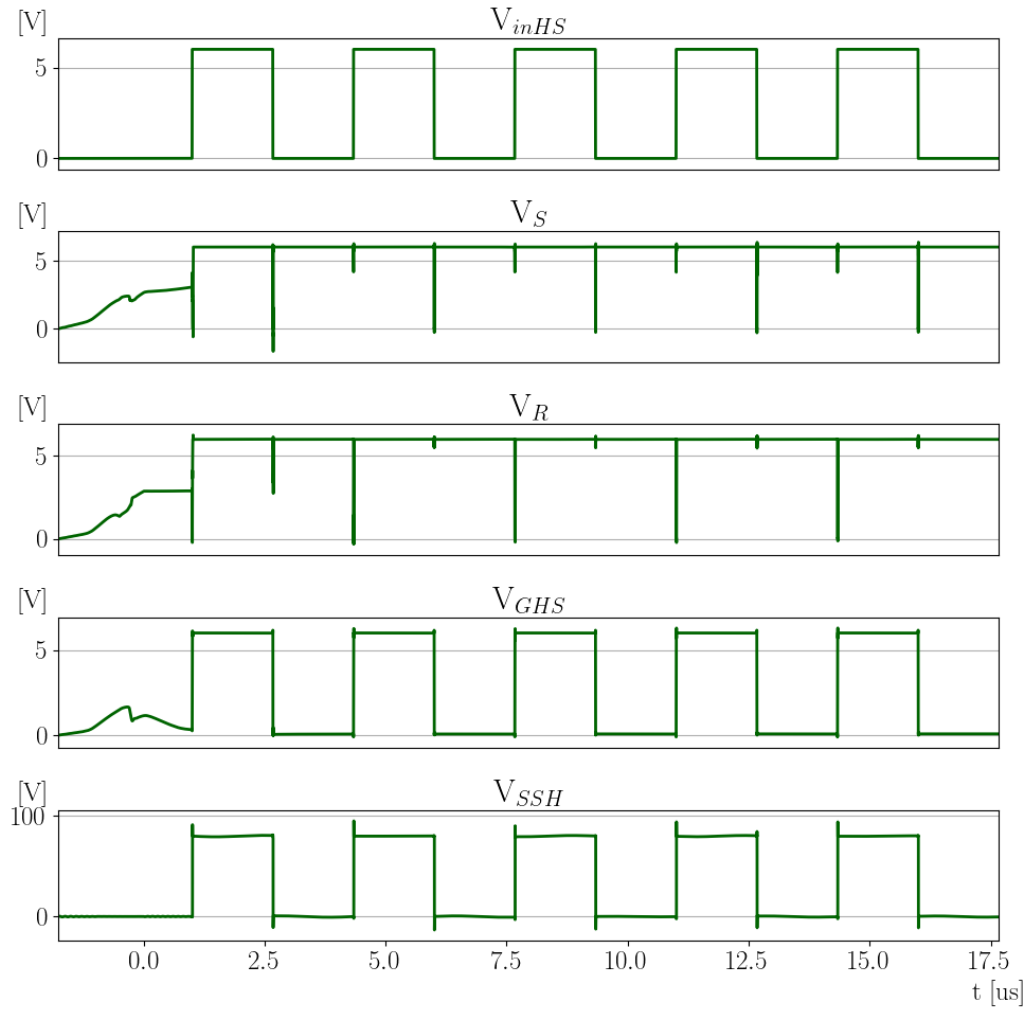


FIGURE 3.7: Transient analysis results for the half-bridge configuration with the first level shifter topology. V_{INHS} and V_{SSH} are referenced to (low-side) ground (0V), while V_S , V_R and V_{GHS} are referenced to V_{SSH} .



FIGURE 3.8: Corner analysis results (voltage V_{GHS}) for the output of the half-bridge configuration with the first level shifter topology, for the nominal case (V_{nom}) and worst-case corners: s , $125^\circ C$ ($V_{s,125}$), and f , $-40^\circ C$ ($V_{f,-40}$).

Operating Condition Check

In this configuration, corner analysis was done using standard PDK models, but also using Operating Condition Check models. OCC is done to confirm that the circuit operates within design specifications, respecting all safety guidelines (e. g. V_{DS} should not exceed the maximum nor fall below the minimum allowed value for any transistor in the circuit).

The results of the corner analysis (using OCC models) for the worst-case corners (f , $-40^\circ C$, and s , $125^\circ C$) are shown in Figure 3.8. Figure 3.8 is a zoom-in to the third operating cycle, since the propagation delays stated in this work were calculated during this cycle. To ensure intended functionality of the level shifter and convergence for all corners, transistor E_2 in the low-side inverters was replaced with three parallel transistors of the same size, leading to increased current consumption of the low-side, but also to reduced propagation delay. In addition, to ensure V_{GS} of the critical transistors in the level-shifting core is between its limits, resistors R_3 - R_6 were added during this check.

Power Density Check

The final check to make before proceeding to the layout design is the power density check. For each transistor, one must manually check whether the power density of the said transistor is low enough (up to 1 W/mm^2 as a Si-based generic rule to prevent thermal runaway). The power density is calculated using Equation 3.2, where $\overline{V_{DS}}$ is the average drain-source voltage of the transistor, $\overline{I_D}$ is the average drain current of the transistor, and A_{est} is the estimated area of the transistor when instantiated in *Layout XL*.

$$\text{Power density} = \frac{\overline{V_{DS}} \overline{I_D}}{A_{est}} \quad (3.2)$$

An important conclusion was drawn during this check - in many instances, *Type 2* inverter presented in Chapter 3 exhibits a power density that is an order of magnitude ($> 10 \text{ W/mm}^2$) higher than the maximum allowed power density. Even if the sizes of transistors were enlarged, this would result in additional current consumption, hence the power density issue would hardly be solved while maintaining a reasonable total current consumption.

3.4 Summary

Understanding the circuit topology and the trade-offs it introduces is crucial to successful IC design. That is why this chapter focused on explaining the level shifter topology based on the work in [1], and the changes needed to map the circuit to the GaN-technology used. A detailed design approach (topology selection, sizing rationale, necessary modifications, etc.) was described - going from the basic building blocks design, such as logic gates design, to the complete system design.

Moreover, fundamental simulations were set up and executed for the first proposed level shifter topology, yielding DC, transient, and corner analysis results presented in this chapter. The half-bridge configuration was introduced and used as well, where the level shifter was simulated with $CMTI > 100 \text{ V/ns}$ for an additional robustness check. More detailed simulation settings may be seen in Appendix C, and the comparison with the second topology, for all important parameters (propagation delay, CMTI, estimated current consumption and area), is given in Chapter 5, Table 5.1.

Chapter 4

Circuit Design -Topology 2

In the following chapter, the second proposed level shifter topology and operation principle are described. Afterwards, the design methodology of the circuit is elaborated - utilizing simple building blocks such as logic gates described in Chapter 3, and building the entire system. In addition, relevant simulation results are presented and discussed - very similar to what was done in Chapter 3, which will be referenced in this chapter quite often.

4.1 Proposed Level Shifter and its Operation Principle

The second proposed level shifter topology based on [2] is shown in Figure 4.1 and is discussed in this section.

As stated in [2] and seen in Figure 4.1, the topology in question utilizes a few techniques to ensure sufficient CMTI and improve mismatch tolerance of the level shifter. **Note:** in the following paragraphs of this section, all the signals, nodes, and components mentioned are the ones seen in Figure 4.1.

The first technique in question - FeedBack Discharge Control (FBDC) - is almost identical to the DDC method described in Chapter 3 and based on [1], hence its operation principle will be omitted here. However, it is worth noting that there is a single difference between FBDC and DDC - namely, diodes (E_7 , E_8) are added at drains of the blanking transistors (E_9 , E_{10}) to "further clamp the V_{DS} of E_9 and limit the discharge current through it, preventing FB oscillation at slower dV/dt " [2]. For more details, see [2].

Furthermore, Common-Mode Blanking Pulse (CMBP) is employed to mitigate the false trigger of the high-side logic in the event of a negative common-mode dV_{SSH}/dt . A pulse V_B is generated (via instance $I16$) to pull V_{N3} (V_{N4}) low by turning on transistor E_9 (E_{10}), and a pulse V_{CM} is generated (via instances $I17$, $I18$, NOR) to prevent any residual, faulty logic states from reaching S/R nodes, leaving the output signal unchanged. For more details, see [2].

Finally, MisMatch Reduction Circuit (MMRC) reduces mismatch effects during positive dV_{SSH}/dt . In a real circuit, symmetrical components (e. g., transistors E_5 and E_6 in Figure 4.1a) which are ideally equal, slightly differ. Therefore, positive

4. CIRCUIT DESIGN -TOPOLOGY 2

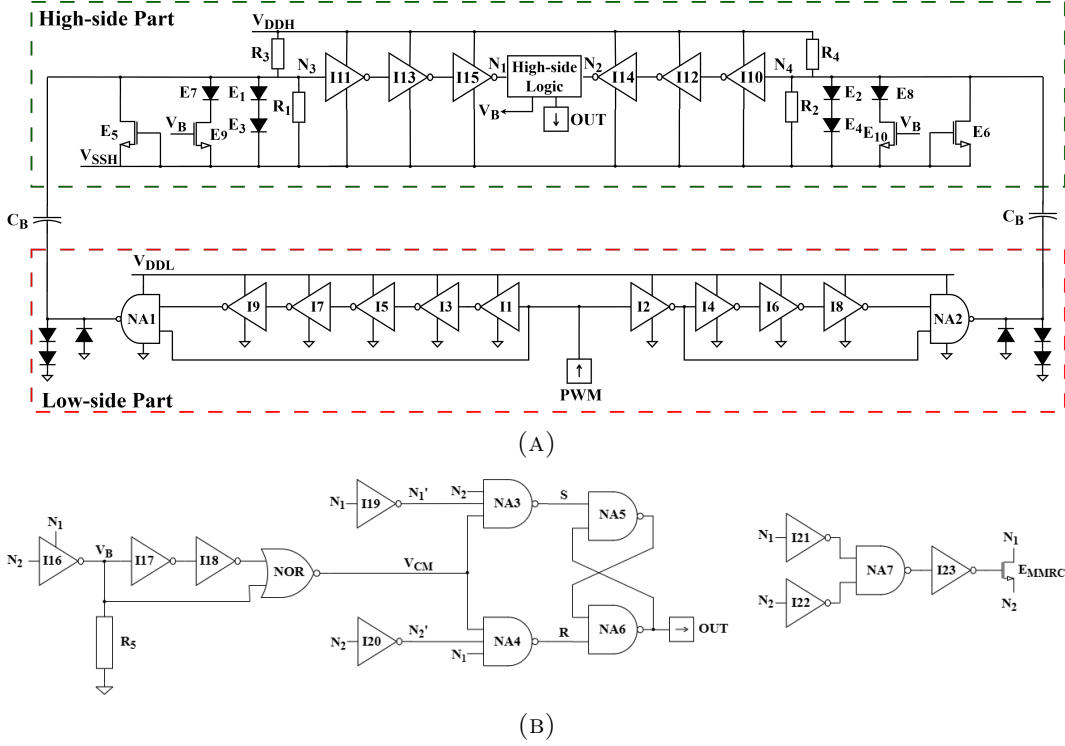


FIGURE 4.1: (A) Capacitive level shifter topology based on [2]. (B) High-side logic comprised of the Common-Mode Blanking Pulse (CMBP), a Set-Reset Flip-Flop (SRFF), and the MisMatch Reduction Circuit (MMRC), respectively.

common-mode pulses at nodes N_3 and N_4 might not occur simultaneously. The circuit may wrongly interpret this difference in the common-mode signals as a differential signal, and the false triggering of the SRFF may occur. Hence, the two different common-mode signals should be equalized before reaching the SRFF to ensure functional circuit operation. This is done by generating a positive voltage pulse at the gate of E_{MMRC} (via instances $I21$, $I22$, $NA7$, $I23$), thus equating voltages at nodes N_1 and N_2 . For more details, see [2].

A few notable changes with respect to the initial topology described in [2] were made and are listed as follows:

- transistors M_3 , M_5 , M_{10} and M_{12} from [2] are omitted: M_3 , M_{10} due to lack of p -type devices, and M_5 , M_{12} because they would be constantly on, pulling nodes N_3 and N_4 low when not needed;
- diodes $E_1 - E_4$ and low-side diodes were added to provide sufficient current path in the event of large dV_{SSH}/dt transitions;
- resistors were added: R_1 and R_2 to ensure pulses V_{N3} and V_{N4} are large enough (exceed sensing inverters' thresholds) to be noticed by sensing inverters and forwarded to the rest of the logic, and R_3 and R_4 to ensure V_{N3} and V_{N4} do not fall out of the breakdown limits (added to pass OCC, as in Chapter 3).

Parameter(s)	Value	Unit
C_B	400	fF
R_1, R_2	70	$k\Omega$
R_3, R_4	60	$k\Omega$
R_5	50	$k\Omega$
$W_1 - W_4$	1000	μm
W_5, W_6	300	μm
$W_7 - W_{10}$	400	μm
W_{MMRC}	400	μm
W_{LS}^*	400	μm

TABLE 4.1: Sizing of the second capacitive level shifter topology. $^*W_{LS}$ refers to widths of the low-side diodes.

4.2 Level Shifter Design

This section is very similar to section 3.3 in Chapter 3. For that reason, the simulation results will mainly be presented in this section.

4.2.1 Sizing the Circuit

In the following subsection, design choices regarding sizing of the components in the second capacitive level shifter topology are discussed. **Note:** for the rest of this chapter, the "level shifter" discussed is the one in Figure 4.1, and in the following paragraphs of this subsection, all the signals, nodes, and components mentioned are the ones seen in Figure 4.1.

The logic gates from the first level shifter topology were reused, and during the OCC and power density check, needed logic gates were resized to pass the two checks. The capacitors were sized referring to [2], to provide sufficient isolation, and were later resized for the level shifter to be able to operate across critical corners. Resistors, diodes and blanking transistors were sized using the identical rationale as explained in Chapter 3, and then modified to ensure proper level shifter operation.

The dimensions of all components, excluding logic gates, are presented in Table 4.1. Note that all transistors are of minimum-length ($L_{min} = 1.3\mu m$), and a detailed sizing of logic gates after the final half-bridge configuration simulations is given in Appendix D.

4.2.2 DC Analysis and Transient Analysis

Similarly to simulating the first level shifter topology, the second level shifter topology was first simulated with a simple capacitive load. DC analysis was performed to confirm that all nodes are at expected voltages (logical one/zero), and the main focus was on transient analysis. The results of the transient analysis for the second level shifter topology may be seen in Figure 4.2. As can be seen, a pulse is transmitted from the Low-side part to the High-side part (nodes N_3, N_4), which

is sensed by a chain of inverters and forwarded to the High-side logic, generating a set (S) or reset (R) pulse to toggle the output accordingly. In the event of a high dV_{SSH}/dt , the output signal stays unchanged.

4.2.3 Corner Analysis

The second level shifter topology was also simulated over the process corners (f , t , s) at -40 , 27 and $125^\circ C$. The results for the worst corners (s , $125^\circ C$ and f , $-40^\circ C$) compared to the nominal case (t , $27^\circ C$) may be seen in Figure 4.3 - as expected, with the rise of temperature, the propagation delay worsens, especially when the slow (s) corner is simulated. Furthermore, it may be seen that the output signal exhibits an undershoot during high positive dV_{SSH}/dt , which is the worst in the case of s , $125^\circ C$. The signal manages to recover from this unwanted behavior in time, but this means that the level shifter topology is not equally reliable across all corners. Figure 4.3 is a zoom-in to the third operating cycle, since the propagation delays stated in this work were calculated during this cycle.

4.2.4 Half-bridge Configuration

The second level shifter topology was put in the same half-bridge configuration as the first one, with the same simulation settings (given in Appendix C). The only difference in the test-benches is that in the case of simulating the second topology, nodes N_3 and N_4 were not cross-coupled because the output is in phase with the input.

The results of the transient analysis are given in Figure 4.4, for a larger number of cycles to ensure the circuit is functional. It may be observed that the level shifter operates as intended - the output signal follows the input signal with a slight delay, and in the event of high dV_{SSH}/dt , the output signal is not affected. There is one unexpected anomaly in the transient behavior of the set (S) node - during high positive dV_{SSH}/dt (> 100 V/ns), it is pulled low, when it should not be. This shows that the design of the second level shifter topology should be further modified to be able to operate properly during high positive dV_{SSH}/dt .

Operating Condition Check

Referring to Chapter 3, OCC was executed in the same fashion for the second level shifter topology.

The results of the corner analysis for the worst-case corners (t , $-40^\circ C$, t , $125^\circ C$) are shown in Figure 4.5. Figure 4.5 is a zoom-in to the third operating cycle, since the propagation delays stated in this work were calculated during this cycle. In addition, as was done for the first level shifter topology, some components of the second topology had to be resized to meet OCC requirements (e. g., the width of diodes E_1 - E_4 had to be multiplied by a factor of 2). However, the topology was still not able to perform as expected across all corners, hence the design of this level shifter should be further reiterated on and improved, which will be addressed in Chapter 5.

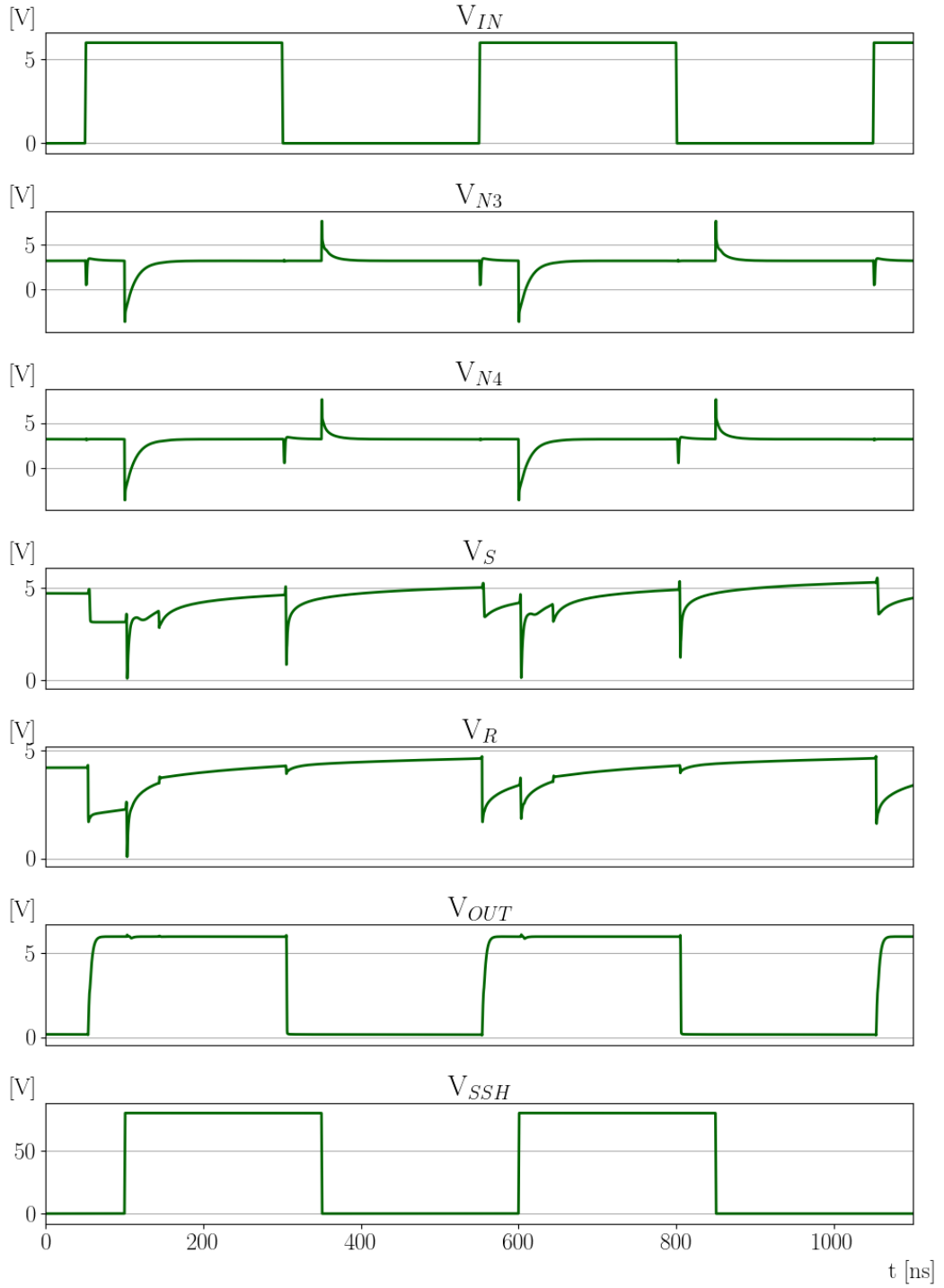


FIGURE 4.2: Transient analysis results for the second capacitive level shifter topology. V_{IN} and V_{SSH} are referenced to (low-side) ground (0V), while V_{N3} , V_{N4} , V_S , V_R and V_{OUT} are referenced to V_{SSH} .

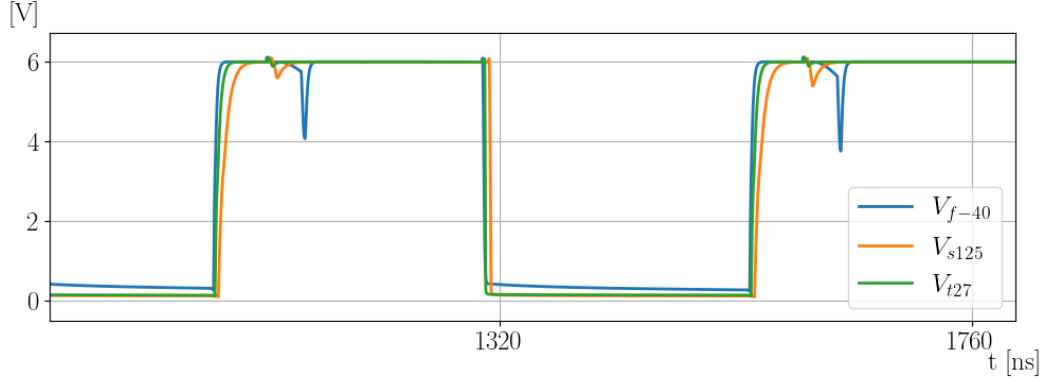


FIGURE 4.3: Corner analysis results for the output of the second capacitive level shifter topology, for the nominal case (V_{nom}) and worst-case corners: s , $125^{\circ}C$ (V_{s125}), and f , $-40^{\circ}C$ (V_{f-40}).

Power Density Check

Finally, as was done for the first proposed level shifter topology, a power density check was performed for the second proposed level shifter topology. For each device, a manual check was performed to confirm whether the power densities of the devices remain within the limit of 1 W/mm^2 . To ensure this condition is satisfied, the majority of logic gates were of *Type 1* previously described - sizing details regarding logic gates are given in Appendix D. For more details on power density calculation, see subsection 3.3.4 in Chapter 3.

4.3 Summary

As there is no single solution to any engineering problem, hence also to the one addressed in this work, a topology different from the one described in Chapter 3 is observed in this chapter. The level shifter topology based on the work in [2] is presented and the changes needed to map the circuit to the GaN-technology used are elaborated. A detailed design approach (topology selection, sizing rationale, necessary modifications, etc.) was described.

Moreover, fundamental simulations were set up and executed for the second proposed level shifter topology, yielding DC, transient, and corner analysis results presented in this chapter. The half-bridge configuration was used as well, where the level shifter was simulated in a real circuit environment for an additional robustness check.

More detailed simulation settings may be seen in Appendix C, and the comparison with the first topology, for all important parameters, is given in Chapter 5, Table 5.1.

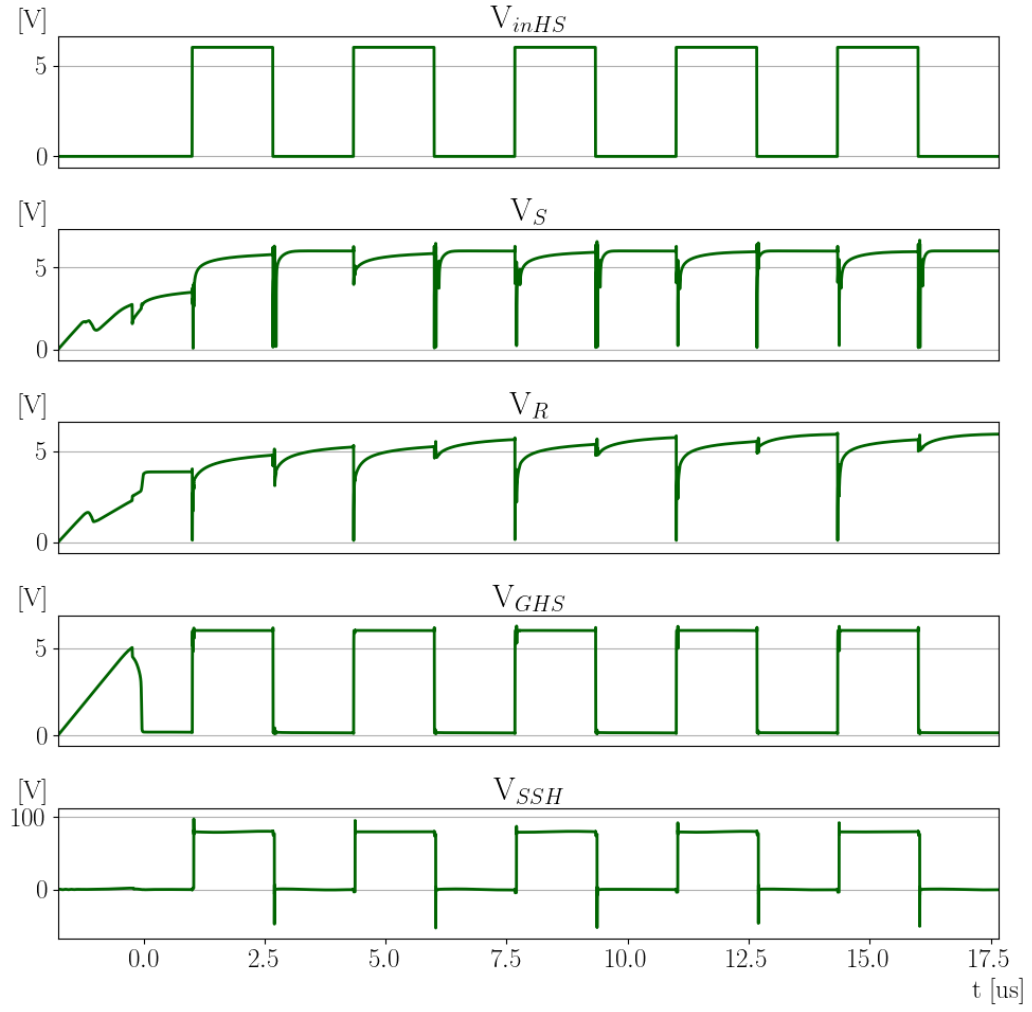


FIGURE 4.4: Transient analysis results for the half-bridge configuration with the second level shifter topology. V_{INHS} and V_{SSH} are referenced to (low-side) ground (0V), while V_S , V_R and V_{GHS} are referenced to V_{SSH} .



FIGURE 4.5: Corner analysis results (voltage V_{GHS}) for the output of the half-bridge configuration with the second level shifter topology, for the nominal case (V_{nom}) and worst-case corners: t , $125^{\circ}C$ (V_{t125}), and t , $-40^{\circ}C$ (V_{t-40}).

Chapter 5

Topology Comparison, Final Remarks and Future Work

5.1 Topology Comparison

The most important parameters - the highest simulated CMTI, the propagation delay, the total current consumption and the estimated area - were extracted from the final simulation (t , $27^\circ C$ in the half-bridge configuration) for each topology, and are given in Table 5.1, the pinnacle of this work.

One may observe that both topologies successfully withstand voltage transients $dV_{SSH}/dt > 100 \text{ V/ns}$ - this is characterized by taking the minimum and maximum values of Equation 5.1.

$$\text{CMTI} = \frac{dV_{SSH}}{dt} \quad (5.1)$$

Moreover, it can be noted that the second level shifter topology has a significantly (nearly $x2$) smaller propagation delay in the case of output rising edge delay, as well as output falling edge delay. This might be counter-intuitive since the second topology utilizes more logic gates, so it is reasonable to assume its propagation delay would be longer than the propagation delay of the first topology. However, the first topology utilizes bigger logic gates and resistors (as resized during OCC and the power density check), which exhibit higher time constants, hence the total propagation delay is increased. This may also be surmised from the power consumption and estimated area - while the first topology consumes somewhat less current than the second topology (due to using fewer components than the second topology), its estimated area is quite close to the estimated area of the second topology (due to using bigger components than the second topology).

5.2 Final Remarks and Future Work

In the end, I would like to briefly address two more matters - which could have been done more efficiently during this thesis, and which is encompassed in future

5. TOPOLOGY COMPARISON, FINAL REMARKS AND FUTURE WORK

Specification	Value (<i>Topology 1</i>)	Value (<i>Topology 2</i>)	Unit
$CMTI_{min}$	-335	-273	V/ns
$CMTI_{max}$	324	300	V/ns
$t_{delay,rise}$	8.46	4.57	ns
$t_{delay,fall}$	9.65	5.57	ns
I_{DC}	8.62	9.43	mA
A_{est}	0.93	0.98	mm^2

TABLE 5.1: Final results for both level shifter topologies extracted from the half-bridge configuration simulations (t , $27^\circ C$).

work on level-shifting circuits.

As stated in 4, the second level shifter topology, when put in the half-bridge configuration and using OCC models, was not functional over all nine corners. This is a consequence of perhaps spending too much time on improving the first topology to pass all needed checks, leaving too little time for the second topology. Moreover, this did not leave enough time to additionally run Monte Carlo simulations and check the mismatch tolerances of both topologies. Hence, work activities and deadlines could have been better managed.

Consequently, future work would entail further improving the second level shifter topology such that it is functional over all corners in the half-bridge configuration, while satisfying OCC and the power density check. Furthermore, mismatch simulations are to be performed and after making sure the topologies are robust enough to mismatch (e. g. 99 out of 100 samples operate properly for 3σ variation), the layout design may commence.

Appendices

Appendix A

GaN Transistors

In the following appendix, more detailed information about certain concepts in Chapter 2 are given.

A.1 Wurtzite Crystal Structure

Figure A.1 shows the wurtzite structure of the crystalline GaN. As stated in [3]:
"Because this structure is very chemically stable, it is mechanically robust and can withstand high temperatures without decomposition. This crystal structure also gives GaN piezoelectric properties that lead to its ability to achieve very high conductivity compared with other semiconductor materials. Piezoelectricity in GaN is predominantly caused by the displacement of charged elements in the crystal lattice. If the lattice is subjected to strain, the deformation will cause a miniscule shift in the atoms in the lattice that generate an electric field – the higher the strain, the greater the electric field."

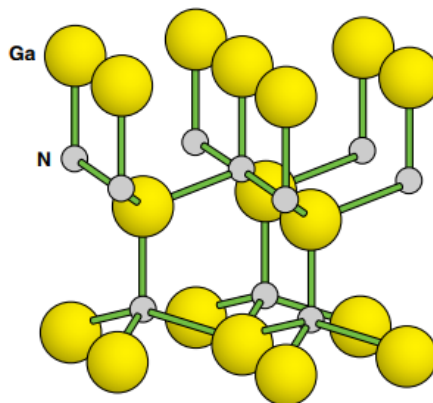


FIGURE A.1: Schematic of wurtzite GaN [3].

Appendix B

Logic Gates

In the following appendix, more detailed information about NAND and NOR logic gates mentioned in Chapter 3 are given.

B.1 NAND-gate

Once the inverter cell has been designed, other logic gates can be easily generated as well. The first among them is a NAND-gate - the schematic is shown in Figure B.1a; sizing is very similar to sizing the *Type 2* inverter - view Appendix B for details.

As is well known, the output of this gate (when there are two inputs) is logical 1 for all combinations of inputs, except when both inputs are logical 1. The results of the transient analysis in Figure B.2 show that the gate implemented indeed works as intended (the settings of the transient analysis are kept the same as for the inverter topologies).

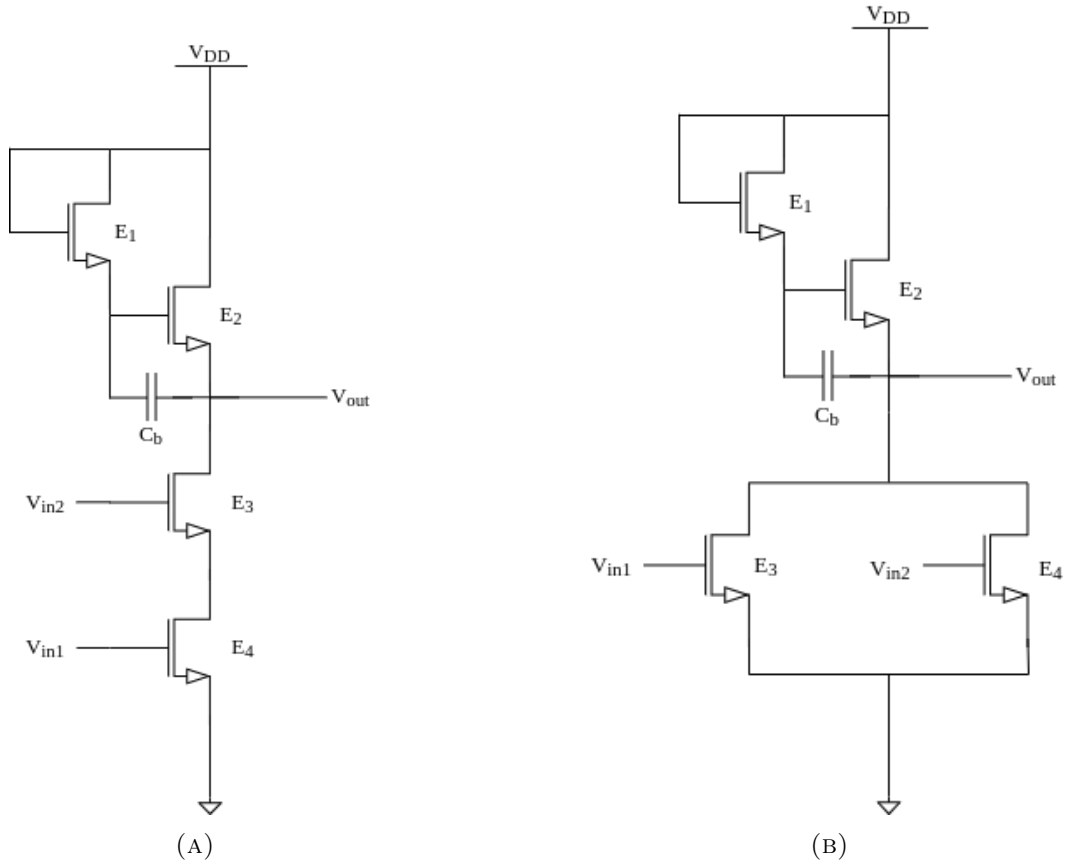
The specifications achieved are listed in Table B.1.

B.2 NOR-gate

Another needed logic gate is the NOR-gate - the schematic is shown in Figure B.1b. The expected behavior of this gate is as follows: for a two-input NOR-gate, output signal needs to be a logical 0 for each combination of inputs, except when both inputs are a logical 1. The transient analysis results in Figure B.3 confirm the correct operation of this logic gate.

The specifications achieved are listed in Table B.1.

All the transistors in both gates are of minimum-length ($L_{min} = 1.3\mu m$), while other parameters are given in Table B.2. The logic gates may have been resized (e. g. $2x$ stronger pull-down network) while executing corner analysis, OCC or power density check - the final values of all the logic gates' sizing parameters (after executing half-bridge configuration simulations) are showcased in Appendix D.


 FIGURE B.1: (A) NAND-gate and (B) NOR-gate topology based on *Type 2* inverter topology [4].

Specification	Value (<i>NAND</i>)	Value (<i>NOR</i>)	Unit
t_{rise}	0.30	1.07	<i>ns</i>
t_{fall}	0.61	0.22	<i>ns</i>
t_{delay}	0.46	0.65	<i>ns</i>
I_{DC}	373.7	401.4	μA
A_{est}	4200	3800	μm

TABLE B.1: Simulation results for NAND- and NOR-gates.

Parameter	Value (<i>NAND</i>)	Value (<i>NOR</i>)	Unit
C_b	25.6	25.6	<i>fF</i>
W_1	40	40	μm
W_2	2	2	μm
W_3	8	4	μm
W_4	8	4	μm

TABLE B.2: Sizing of NAND- and NOR-gates.

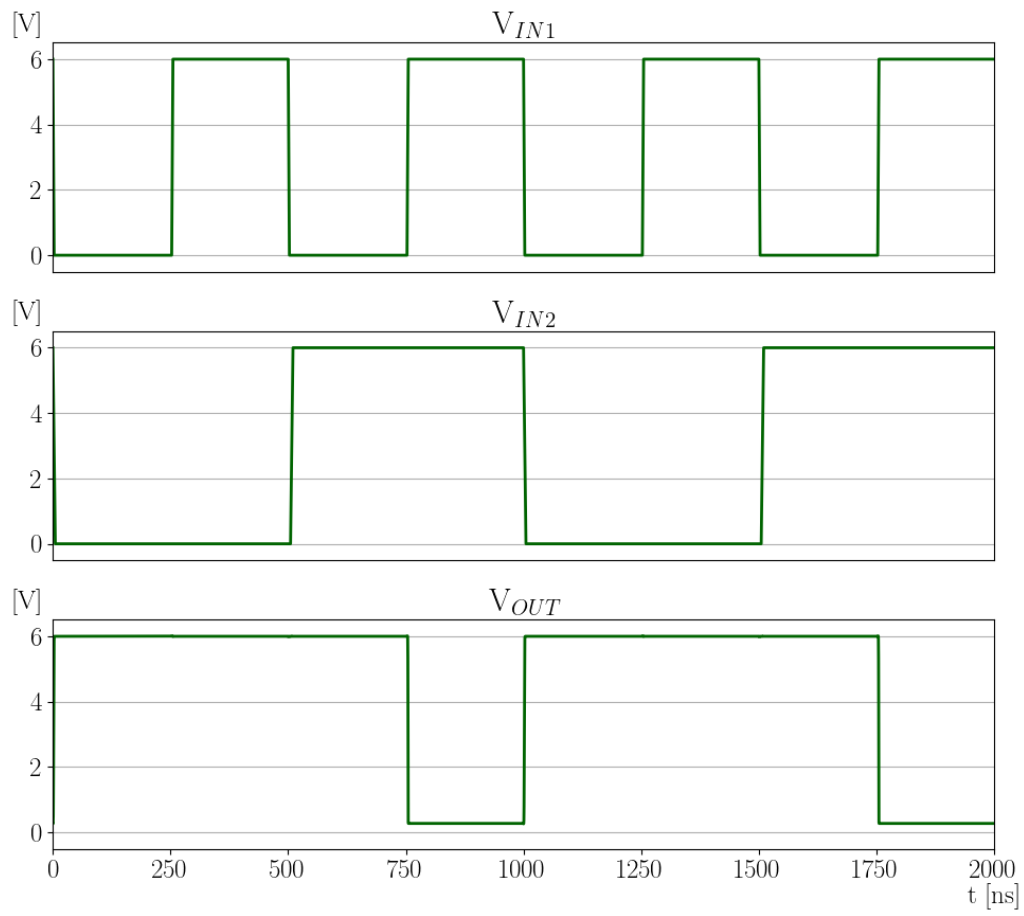


FIGURE B.2: Transient analysis results for the NAND-gate.

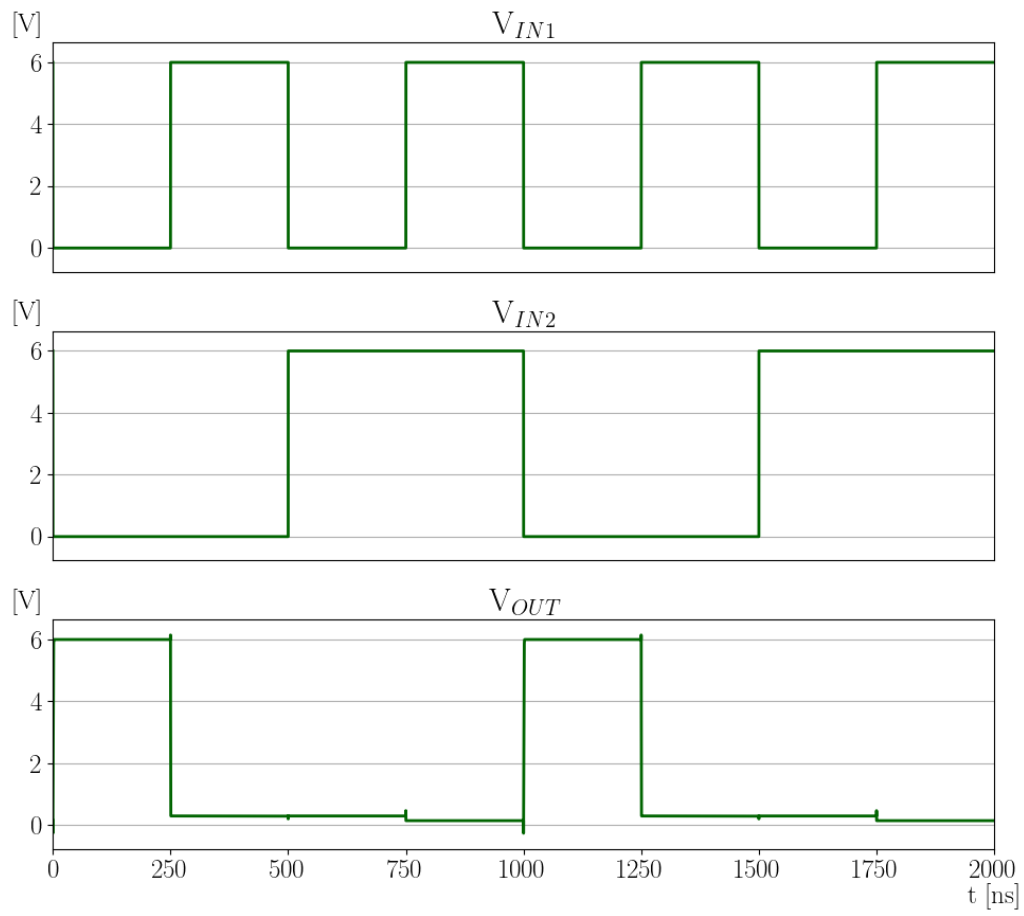


FIGURE B.3: Transient analysis results for the NOR-gate.

Appendix C

Circuit Simulations Setup

In this appendix, test-bench environments for level shifter simulations mentioned in Chapters 3 and 4 are described in more detail.

C.1 DC, Transient and Corner Analysis

The fundamental simulation settings were adjusted as follows:

- *Setup => High Performance Simulation Options => Simulation Performance Mode = **APS**;*
- *Simulation => Options => Analog => $iabstol = 1e-9$, high voltage = **yes**;*
- *Outputs => Save all... => Select signals to output (save) = **all**, Select device currents (currents) = **selected**.*

The test-bench used for the DC, transient and corner analysis is shown in Figure C.1, and the settings for each simulation are as noted:

- DC: *Save DC operating point => **yes**;*
- transient: *Stop time = $1.1 * VAR("nb_cycles") / VAR("f_sw")$, Accuracy defaults = **conservative**;*
- corner: *Model Groups = (**t, f, s**), Temperature = (- 40, 27, 125) °C.*

Voltage signals and load capacitance were set as follows:

- V_{DDL} : $vdc = 6 V$;
- V_{SSH} : *vpulse 0 - 80 V, at $f_{sw} = 2 MHz$, Period = $1/f_{sw}$, Delay time = Period/5, Rise time = Fall time = 0.8 ns (for 100 V/ns);*
- V_{DDH} : $V_{SSH} + (vdc = 6 V)$;
- V_{IN} : *vpulse 0 - 6 V, at $f_{sw} = 2 MHz$, Period = $1/f_{sw}$, Delay time = Period/10, Rise time = Fall time = 1 ps;*
- $C_L = 70 fF$.

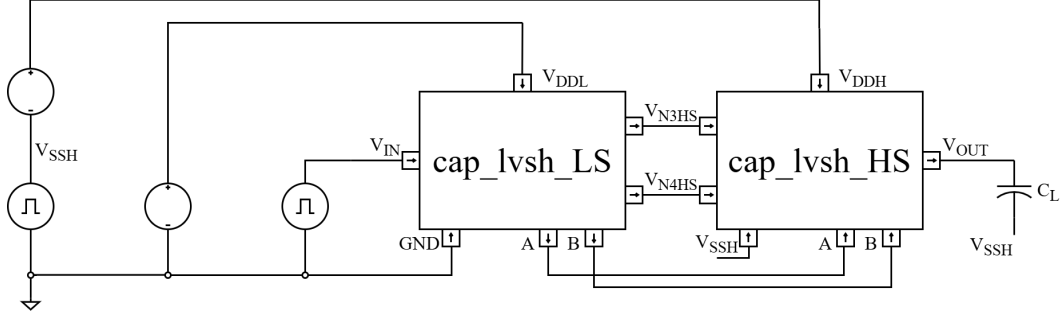


FIGURE C.1: Schematic of the level shifter test-bench for the DC, transient and corner analysis.

C.2 Half-bridge Configuration

The fundamental simulation settings were adjusted as described in the previous section, with slightly changed transient simulation settings:

- transient: *Stop time* = `VAR("td_sw") + VAR("nb_cycles")/VAR("fsw")`, *Accuracy defaults* = **conservative**, *Options => Time Step => minstep* = **1p**, *Algorithm => skipdc* = **sigrampup**, *cmin* = **0.01f**, *maxiters* = **5**;
- $td_{sw} = 1 \mu s$, $nb_cycles = 5$, $f_{sw} = 300 kHz$.

The test-bench used is the one described in Chapter 3 (shown in Figure 3.6). Half-bridge components are configured as follows:

- $L_B = 1 nH$;
- $ME_{1,2}$: $L_{min} = 1.3 \mu m$, $WF = 100 \mu m$, $NF = 20$;
- $L_L = 100 uH$, $C_L = 1 nF$, $R_L = 10 k\Omega$.

Voltage signals were set as follows:

- V_{DDLs} : $vdc = 6 V$;
- V_{DDHS} : $V_{SW} + (vdc = 6 V)$;
- V_{INLS} : *vpulse* 6 - 0 V, at $f_{sw} = 300 kHz$, *Period* = $1/f_{sw}$, *Delay time* = $td_{sw} = 1 \mu s$, *Rise time* = *Fall time* = $10 ns$;
- V_{INHS} : *vpulse* 0 - 6 V, at $f_{sw} = 300 kHz$, *Period* = $1/f_{sw}$, *Delay time* = $td_{sw} = 1 \mu s$, *Rise time* = *Fall time* = $1 ns$;
- V_{BUS} : $vdc = 80 V$.
- *vcvs*: *Voltage gain* = 1, *Delay time* = $32 ns$, $28 ns$ for Topologies 1, 2, respectively.

Appendix D

Level Shifter Sizing Details

D.1 Topology 1

In this section, detailed sizing of the logic gates used in the first level shifter topology is given. Each instance mentioned is the one seen in Figure 3.1, and the components of each instance correspond to the ones mentioned in either of Figures 3.2, B.1, depending on which logic gate is in question. Gates *NOR*, *NAND3*, *NAND4* are an exception - they were derived from the *Type 1* inverter, hence they will have a resistor R in the pull-up network, instead of components E_1, E_2, C_b from Figure B.1. All transistors are of minimum length ($L_{min} = 1.3 \mu m$).

- $I1, I2, I9, I10, I12, I14$: $W_1 = 80 \mu m, R = 5 k\Omega$;
- $I3, I4, I7, I8, I11, I13$: $W_1 = 40 \mu m, W_2 = 2 \mu m, W_3 = 16 \mu m, C_b = 25.6 fF$;
- $I5, I6$: $R = 20 k\Omega, W_1 = 20 \mu m$;
- *NOR*: $W_3 = W_4 = 200 \mu m, R = 10 k\Omega$;
- *NAND1, NAND2*: $W_1 = 40 \mu m, W_2 = 2 \mu m, W_3 = W_4 = 32 \mu m, C_b = 25.6 fF$;
- *NAND3, NAND4*: $W_3 = W_4 = 150 \mu m, R = 3 k\Omega$.

D.2 Topology 2

In this section, detailed sizing of the logic gates used in the second level shifter topology is given. Each instance mentioned is the one seen in Figure 4.1, and the components of each instance correspond to the ones mentioned in either of Figures 3.2, B.1, depending on which logic gate is in question. Gates *NOR*, *NA1*, *NA2*, *NA5*, *NA6* are an exception - they were derived from the *Type 1* inverter, hence they will have a resistor R in the pull-up network, instead of components E_1, E_2, C_b from Figure B.1. All transistors are of minimum length ($L_{min} = 1.3 \mu m$).

D. LEVEL SHIFTER SIZING DETAILS

- *I1- I9, I12 - I16, I19, I20*: $W_1 = 80 \mu m$, $R = 5 k\Omega$;
- *I17*: $W_1 = 40 \mu m$, $W_2 = 2 \mu m$, $W_3 = 4 \mu m$, $C_b = 25.6 fF$;
- *I10, I11, I18, I21 - I23*: $R = 20 k\Omega$, $W_1 = 20 \mu m$;
- *NOR*: $W_3 = W_4 = 200 \mu m$, $R = 10 k\Omega$;
- *NA1, NA2*: $W_3 = W_4 = 150 \mu m$, $R = 3 k\Omega$;
- *NA3, NA4*: $W_1 = 40 \mu m$, $W_2 = 2 \mu m$, $W_3 = W_4 = W_5 = 32 \mu m$, $C_b = 25.6 fF$;
- *NA5, NA6*: $W_3 = W_4 = 10 \mu m$, $R = 30 k\Omega$.

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